

# David A. Plappert

Principal Cloud Architect · Consulting Architect · AWS Reliability Lead

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Schedule 30 min →

US Patent 11,146,689, Cloud Architecture (2021) · 2× AWS City on a Cloud award (2016, 2018) · 19 years on AWS, since EC2 beta

## SUMMARY

- Principal-level cloud architect with 19 years designing, building, and operating mission-critical AWS platforms; patent holder; two-time AWS innovation award winner
- Consulting-native: 10 years delivering parallel client engagements at a government health-tech consultancy — architecture I scoped and delivered grew one engagement by \$1M/yr and another by \$2M/yr; 18F (federal digital services, GSA) advisor
- Currently the sole Caterpillar-side AWS authority on **Vertex 3D**, Caterpillar's vendor-built real-time 3D platform powering aftermarket commerce — independent architecture review of an eight-figure vendor platform relationship
- Ships AI-augmented engineering in production today: Claude Code orchestrated CI/CD, agentic deployment workflows, MCP-based tool integration, multi-agent parallel orchestration
- Specializes in architecture reviews aligned to the AWS Well-Architected Framework, multi-account AWS architecture, high-availability and disaster-recovery design, observability, platform engineering, and migration strategy
- **Industry depth:** Insurance, Payments, and Finance (10 yrs on state health insurance exchanges, \$500M/yr atomic payments, IRS / Stripe / EDI integrations); Manufacturing (Caterpillar aftermarket-commerce backbone); Healthcare and Federal Government (DCHBX, MA HCB, CoverME, DC DBH, 18F advisor); Retail and Point-of-Sale (Best Buy, billion-dollar BestBuy.com and POS systems on AWS); Media / Streaming (PBS video platform; RTMP / SRT ingest pipelines)

## PROFESSIONAL EXPERIENCE

### Caterpillar Inc. (via TEKsystems), Lead Cloud Engineer / AWS Reliability Lead

September 2025 – Present · Chillicothe, IL

- Sole Caterpillar-side AWS authority for **Vertex 3D**, Caterpillar's real-time 3D collaboration platform powering aftermarket commerce — ~\$3,000/minute in revenue when running, ~30 downstream publisher integrations; technical oversight of an **eight-figure vendor platform relationship**
- Independent architecture review authority: every monthly vendor-shipped AWS CDK release passes my architecture review and pre-deploy testing before production; typically remediate 2–3 defects per release before they can reach the platform
- Author of the technical findings behind Caterpillar's formal architecture-correctness escalations to the vendor; present findings and corrective requirements to Caterpillar business and engineering leadership
- Architected the platform's load-testing, monitoring, and alerting design; supported scaling from 2,000 to 8,000 concurrent users
- Designed and built a Claude Code agentic workflow that runs production deployments, pre-production dry-runs, automated bug detection, ticket filing, and end-to-end CI/CD pipeline orchestration — parallel multi-agent runs against several stacks, bridged into GitHub, AWS CLI, and the team's ticketing system through custom MCP servers and tool definitions I wrote
- Cross-team escalation authority across both AWS and Node.js stacks for Caterpillar's Publisher Distribution Team (PDT)
- Technical lead for Caterpillar's Node.js workstream on Vertex 3D; built and operate multiple production applications on the platform; mentoring junior Node.js engineer
- **Stack:** AWS (Fargate, ECS, S3, RDS / Aurora Serverless, Lambda, CloudWatch, IAM), AWS CDK, CloudFormation, Node.js, Java / JVM, GitHub Actions, Claude Code, Anthropic SDK, MCP

## IdeaCrew (government health-tech consultancy), Principal Cloud Architect & DevOps Lead

2015 – 2025 · Remote

- Employee #3; led cloud architecture and DevOps across parallel client engagements worth >\$50M annually as the firm grew past 100 employees
- Client-facing lead architect for four state-government platforms — DC Health Benefit Exchange (DCHBX), Massachusetts Health Connector for Business, CoverME (Maine), and DC Department of Behavioral Health — owning architecture, security, roadmap, and delivery end to end
- Owned the AWS architecture that won **2x AWS City on a Cloud awards (2016, 2018)** on the DCHBX platform
- Grew engagements through delivery: scoped and solo-built a data warehouse spanning 8 databases (MongoDB BI Connector, MySQL, PostgreSQL) feeding Tableau — **+\$1M/yr contract expansion**; designed 300+ automated HIPAA- and SOC 2-compliant data jobs — **+\$2M/yr additional expansion**
- **18F (federal digital services, GSA) advisor** — AWS architecture consultation and workshops for multiple 18F chapters on state-government health exchange replication
- Migration strategy leadership: led multi-year state health insurance migrations from federal FedHub to state-based exchanges, worth tens of millions; team held the fastest FedHub transition record
- Multi-account AWS governance: managed \$1M/month cloud spend across 12 AWS and 3 Azure accounts with account-level isolation and cost optimization
- Designed distributed data consistency patterns across multi-region, high-availability state-exchange architecture — idempotent EDI processing, eventual-consistency reconciliation, and atomic payment workflows handling >\$500M annually under HIPAA, SOC 2, and IRS Publication 1075 compliance
- Led SOC 2 certification and annual audit processes, including disaster-recovery and business-continuity design; presented architecture decisions and audit outcomes to state-agency executives and client leadership
- Led monolith-to-microservices migration onto AWS EKS, ECS, and Lambda; company-wide final escalation point for production incidents across every client engagement
- **Stack:** Ruby on Rails, Node.js, Angular, AWS, Azure, EKS, ECS, Docker, Terraform, MongoDB, MySQL (Aurora Serverless), DynamoDB, PostgreSQL, Oracle

## VentureBeat, Senior Web Developer / Cloud DevOps

June 2014 – October 2015

- Managed all of VentureBeat's AWS infrastructure (EC2, EBS, Route 53, ELB, S3, CloudFront) with Chef; built CI workflow on GitHub + Jenkins; rolled out New Relic monitoring across every EC2 instance
- Launched **VentureBeat Insight**, a new e-commerce research product that generated **\$2M revenue in its first year**, on a three-person team
- Helped scale venturebeat.com — one of the web's largest WordPress installations — through ~150% year-over-year traffic growth
- Increased newsletter signups 500% through Optimizely experimentation with the marketing team

## PBS, Cloud Engineer

2014

- Built and operated AWS-based RTMP video streaming platform with 99.9% uptime
- Implemented adaptive bandwidth modulation, materially increasing viewer watch time
- Managed a development team of two direct reports

## Argonne National Laboratory, Senior Cloud Architect

2013

- Built and managed the application suite administering hundreds of millions of dollars in competitive federal research funding; grew the lab's funding **152% year-over-year**
- Built dev / stage / prod environments on the lab's OpenStack infrastructure; mentored a college intern to full solo ownership of the application suite

## Best Buy, Senior Cloud DevOps

2012 – 2013

- Owned cloud infrastructure across all Best Buy IT projects — **BestBuy.com and Point-of-Sale, billion-dollar revenue systems** — on AWS (EC2, S3, Route 53, VPC, Auto Scaling Groups, ELB, Direct Connect), automated with Chef
- **Thanksgiving Day 2012, Best Buy received ~25% of all US e-commerce traffic — sole on-site engineer managing and scaling production**; fleet peaked at 1,000+ servers over the holiday season
- Built production services in **Ruby on Rails** and operated **Java / JVM** workloads across the BestBuy.com and Point-of-Sale stacks, with Cassandra in the BestBuy.com data layer
- Reduced infrastructure costs by over 50% while improving deployment speed; led a 6-member DevOps engineering team

## Earlier roles, web & cloud infrastructure engineering

2005 – 2012

- **White House Custom Colour** (2011) — AWS cloud administration; designed MySQL high availability with dual master-replica pairs and automated monitoring-driven failover; rebuilt e-commerce checkout
- **Imagewerks Marketing / CATCO** (2011 – 2012) — mobile web team lead; built a generator that cut mobile-site builds from ~150 hours to under 1 hour
- **Acesse** (2012) — high-traffic PHP / MySQL ad-serving on AWS (EC2, RDS, ElastiCache) with AMI-based auto scaling
- **Apostolic Bible Institute** (2006 – 2010) — Director of Capital Investments, I.T.: campus-wide wireless (20 access points, 6 buildings, 40 acres), live streaming for classes and services, college web platform; first production AWS work during the EC2 beta era — plus part-time I.T. advisory for **Comprehensive Research Group** (2005 – 2012)

## EDUCATION & CREDENTIALS

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- **US Patent 11,146,689**, Cloud Architecture (Issued October 12, 2021)
- **AWS City on a Cloud Award**, 2016 and 2018
- **AWS Certified Solutions Architect – Associate** (2019 – 2021)
- **18F advisor** (federal digital services / GSA), AWS architecture consultation during IdeaCrew tenure
- **19 years on AWS**, operating production workloads since EC2 beta
- **Education**: College coursework; self-taught engineer, production-trained on AWS since 2007

## TECHNICAL SKILLS

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- **Cloud & Infrastructure**: AWS (Fargate, ECS, EKS, S3, RDS, Lambda, CloudWatch, IAM, VPC, Direct Connect, ElastiCache, CDK, CloudFormation), multi-account architecture (AWS Organizations), Azure, GCP
- **Architecture & Reliability**: AWS Well-Architected Framework, distributed systems, event-driven architecture, microservices, multi-region, high availability, disaster recovery / business continuity, hybrid cloud, platform engineering, architecture and design reviews
- **Consulting & Client Leadership**: engagement scoping and expansion, pre-sales technical support, executive workshops and design reviews, multi-client parallel delivery, vendor architecture oversight, mentoring
- **AI Engineering & Agent Tooling**: Claude Code CLI for production engineering work, Anthropic API / SDK, multi-agent parallel orchestration, context engineering, MCP servers and custom tool definitions, AI-assisted CDK / infrastructure development, evaluation harnesses, RAG patterns
- **CI / CD & Developer Productivity**: GitHub Actions, Jenkins, AWS CodePipeline, Chef, agentic CI/CD orchestration, automated PR review, agent-assisted incident analysis and runbook automation
- **AI-SDLC Governance**: IP protection patterns, secret scanning, open-source license compliance, audit trail design
- **Infrastructure as Code & Containers**: AWS CDK, CloudFormation, Terraform, Docker, Kubernetes, EKS
- **Languages & Frameworks**: Ruby / Rails, Node.js / Express, JavaScript / TypeScript (React, Vue / Nuxt, Angular, Astro), Python, Bash, SQL, PHP (author); Java / JVM (operate on AWS); Serverless Framework
- **Databases**: PostgreSQL, MySQL / Aurora Serverless, DynamoDB, Cassandra, MongoDB, Oracle, Redis
- **Observability & Streaming**: New Relic (APM, tracing, alerting), CloudWatch, OpenTelemetry; RTMP / SRT live ingest and adaptive-bitrate pipelines on AWS
- **Compliance & Security**: HIPAA, SOC 2 certification and audit, IRS Publication 1075, security automation